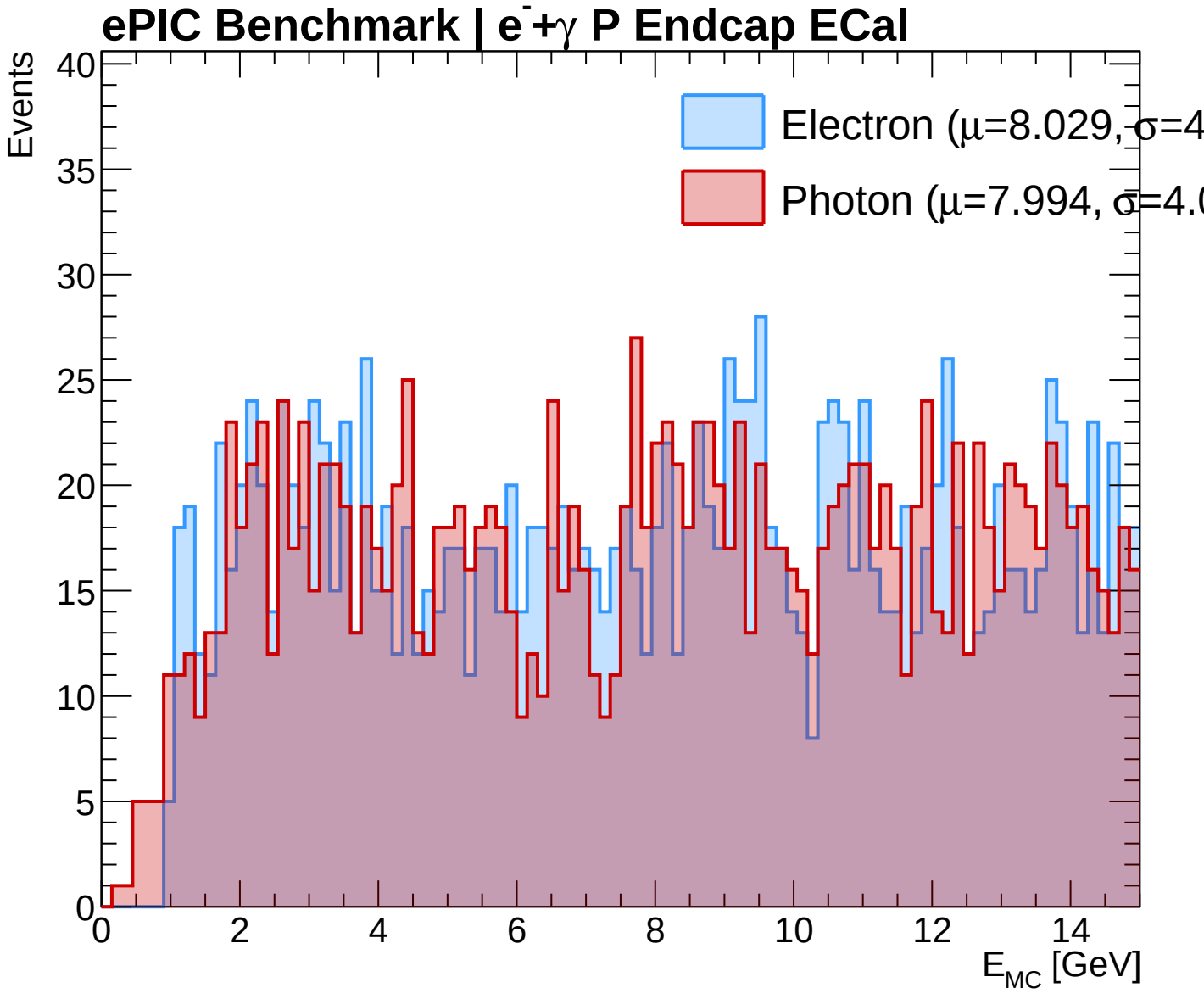
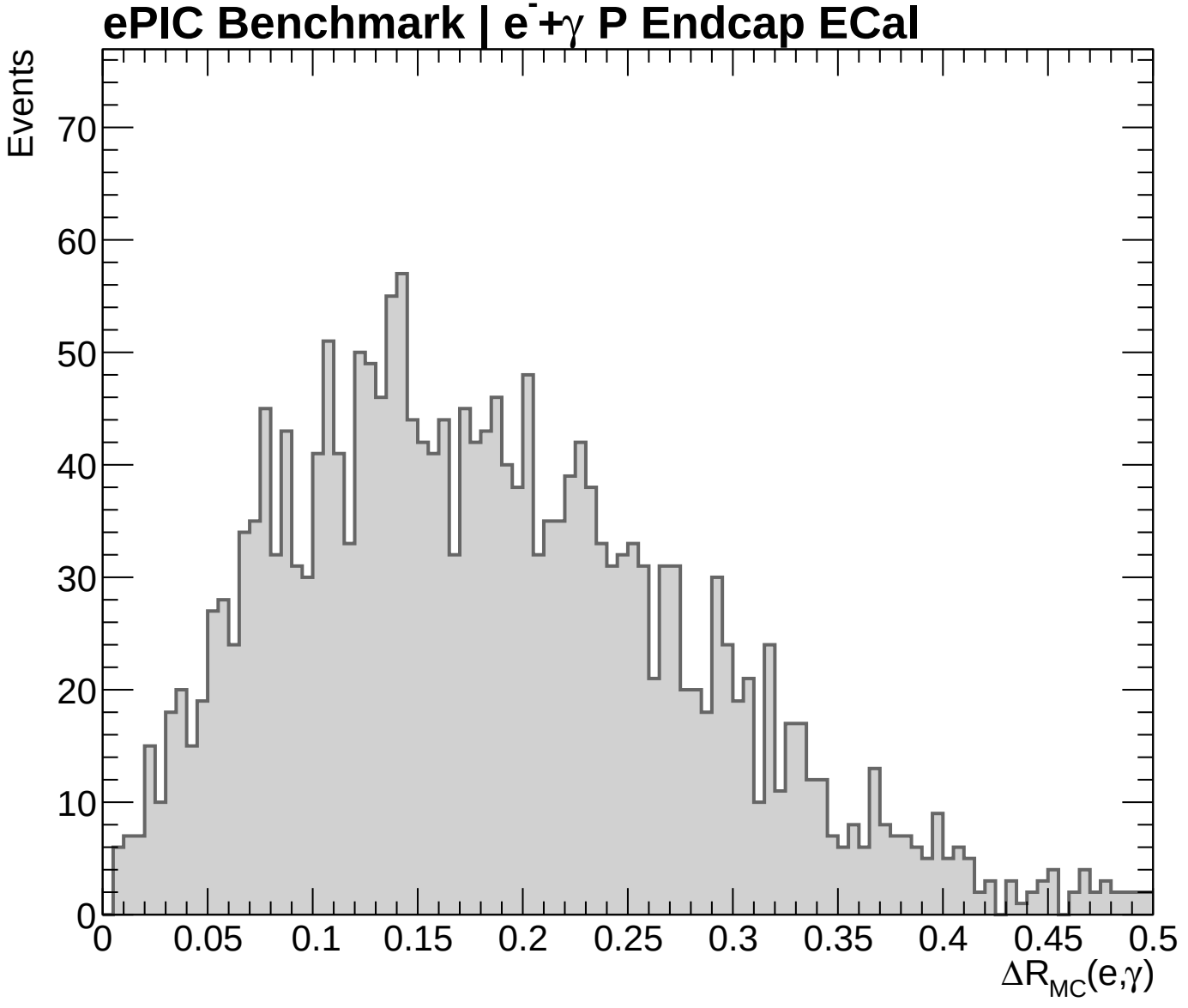
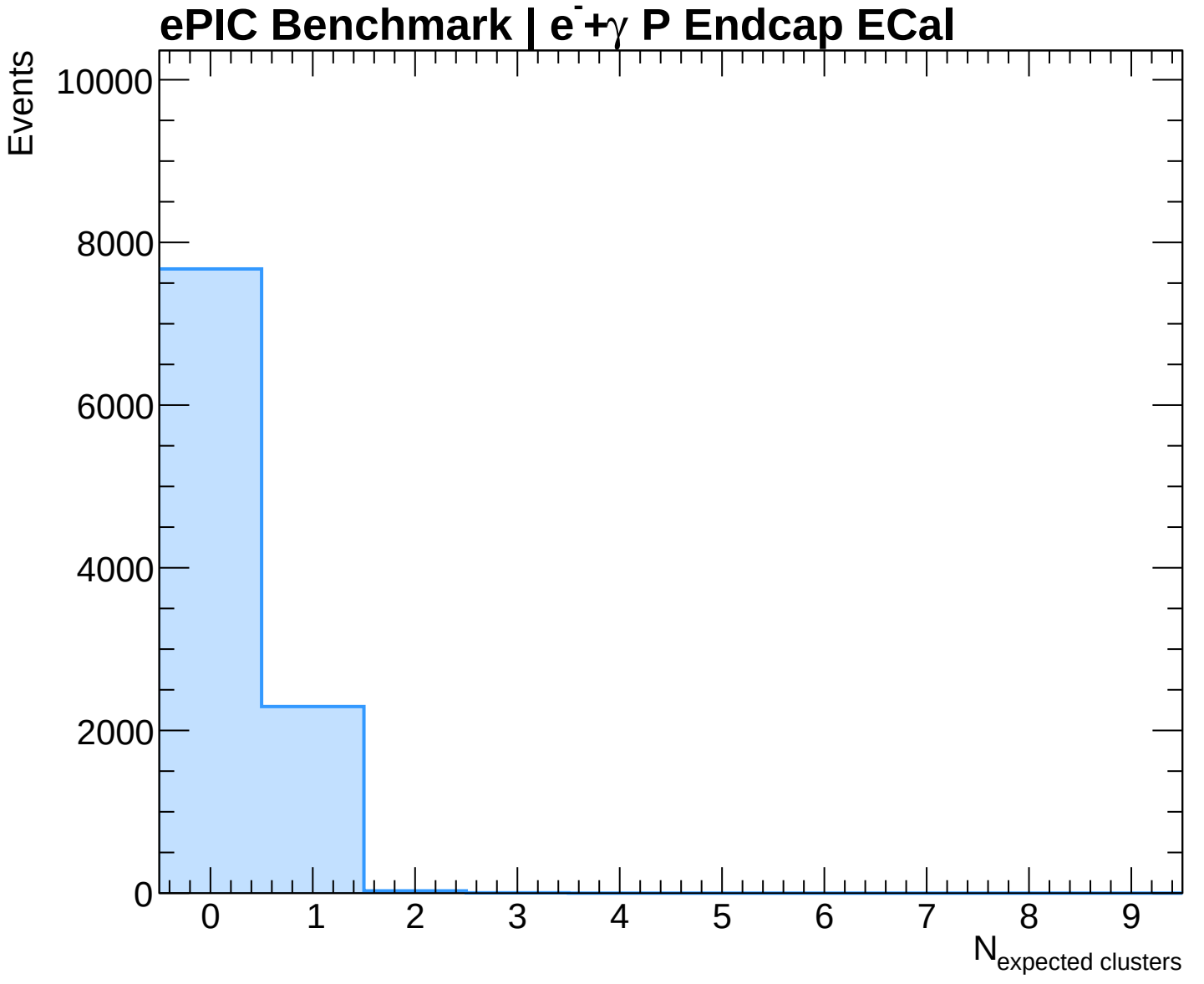
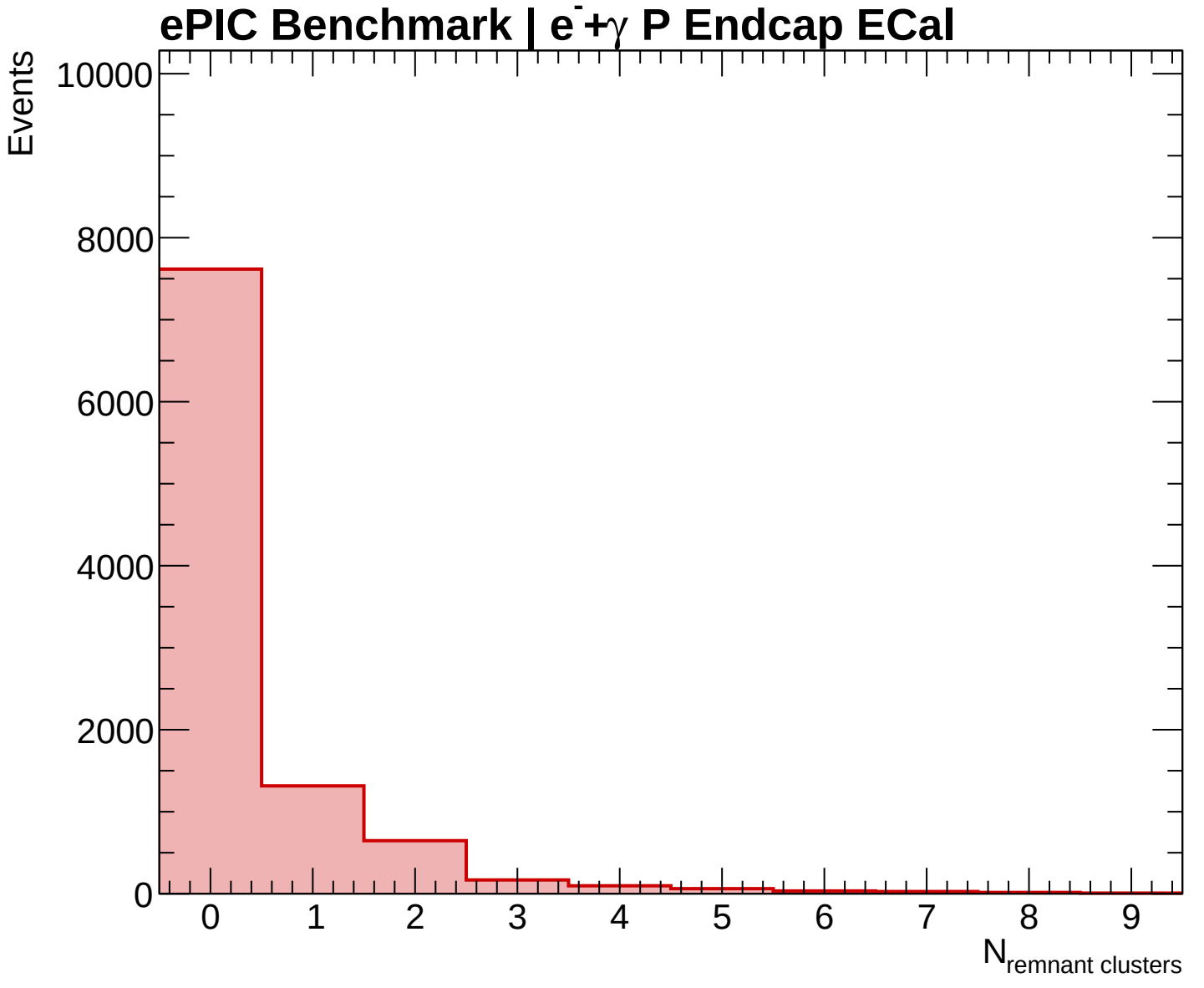
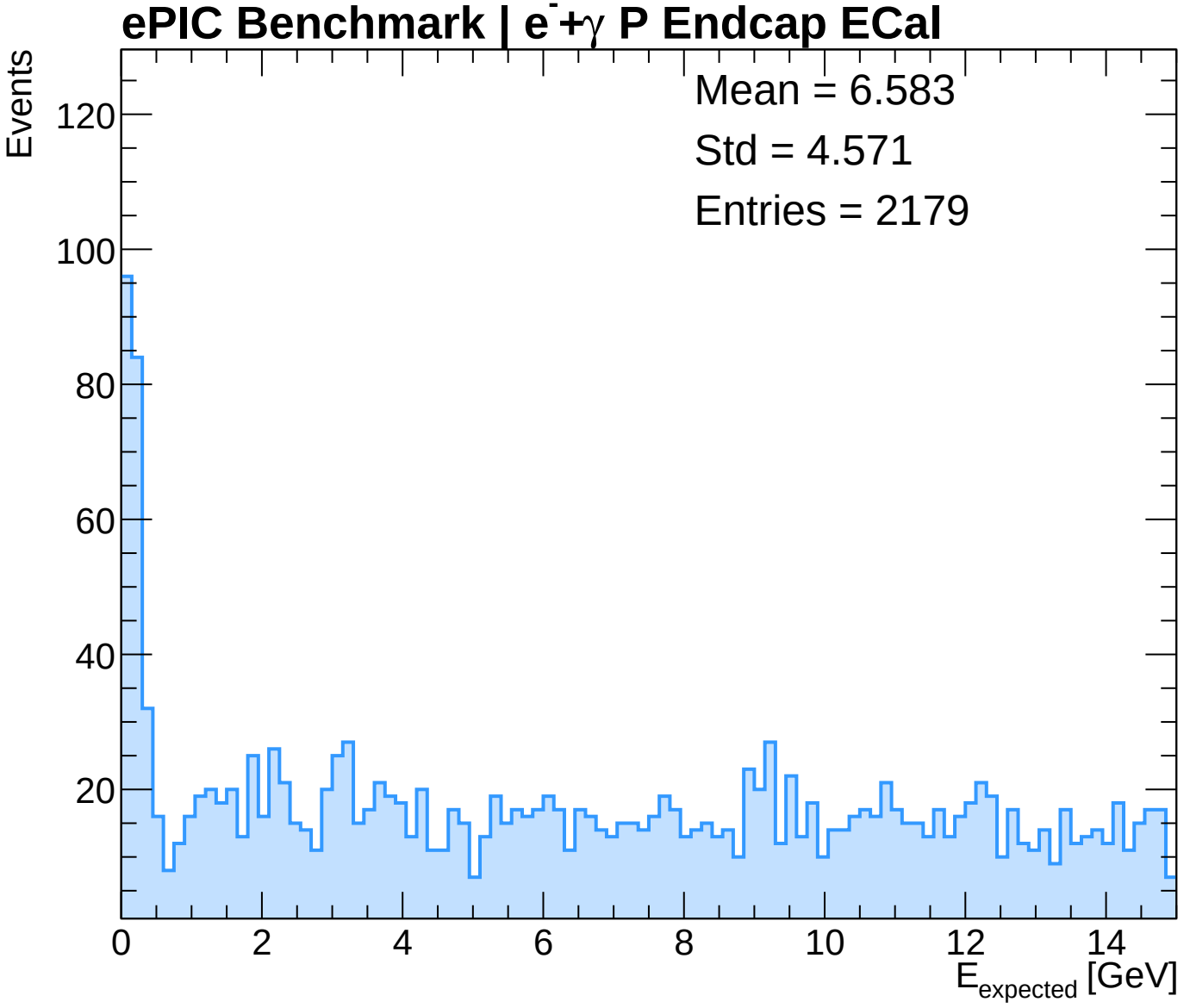
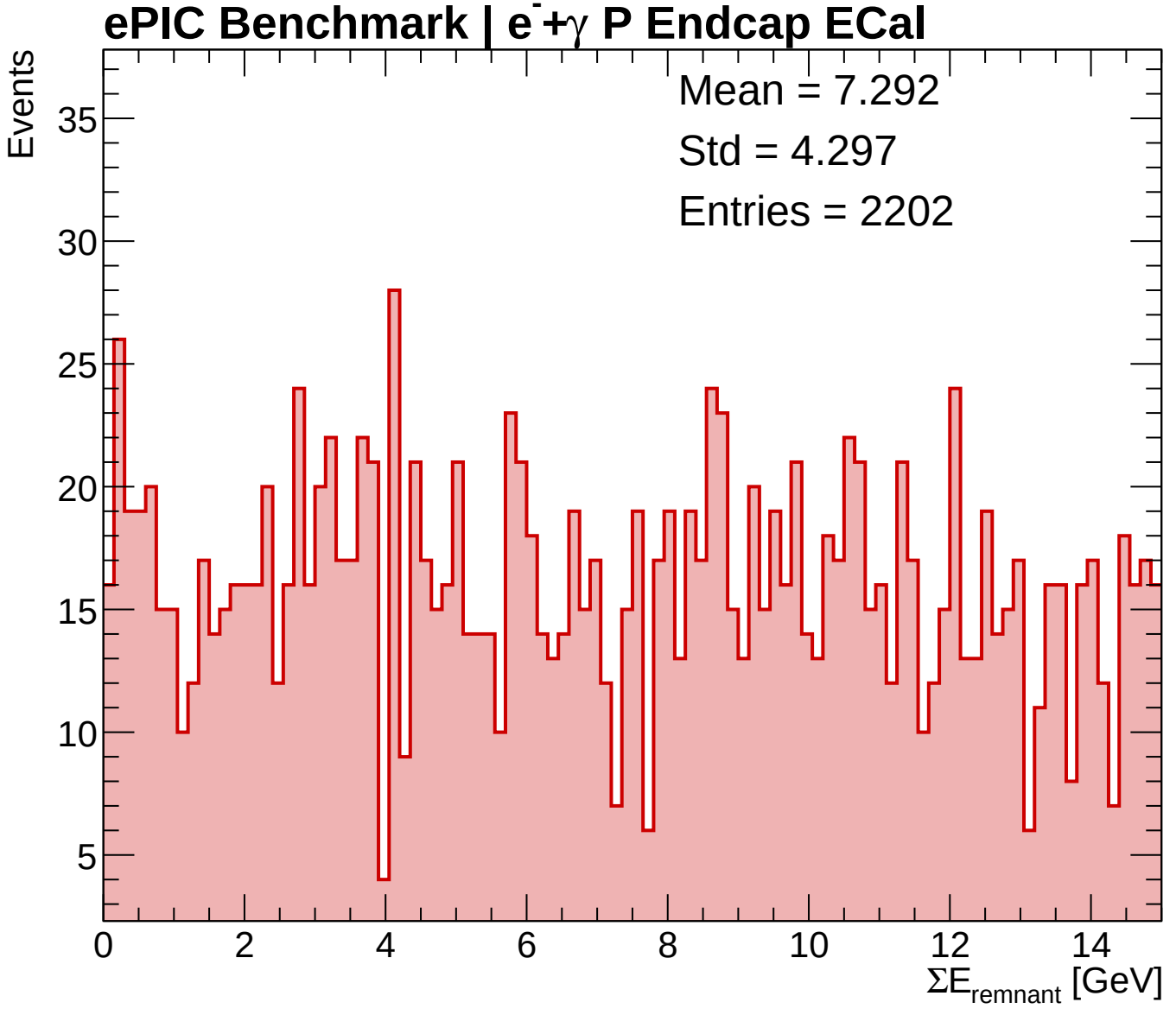

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Electron (μ=8.029, σ=4
Photon (μ=7.994, σ=4.0
0
5
10
15
20
25
30
35
40
0
2
4
6
8
10
12
14
E_MC [GeV]

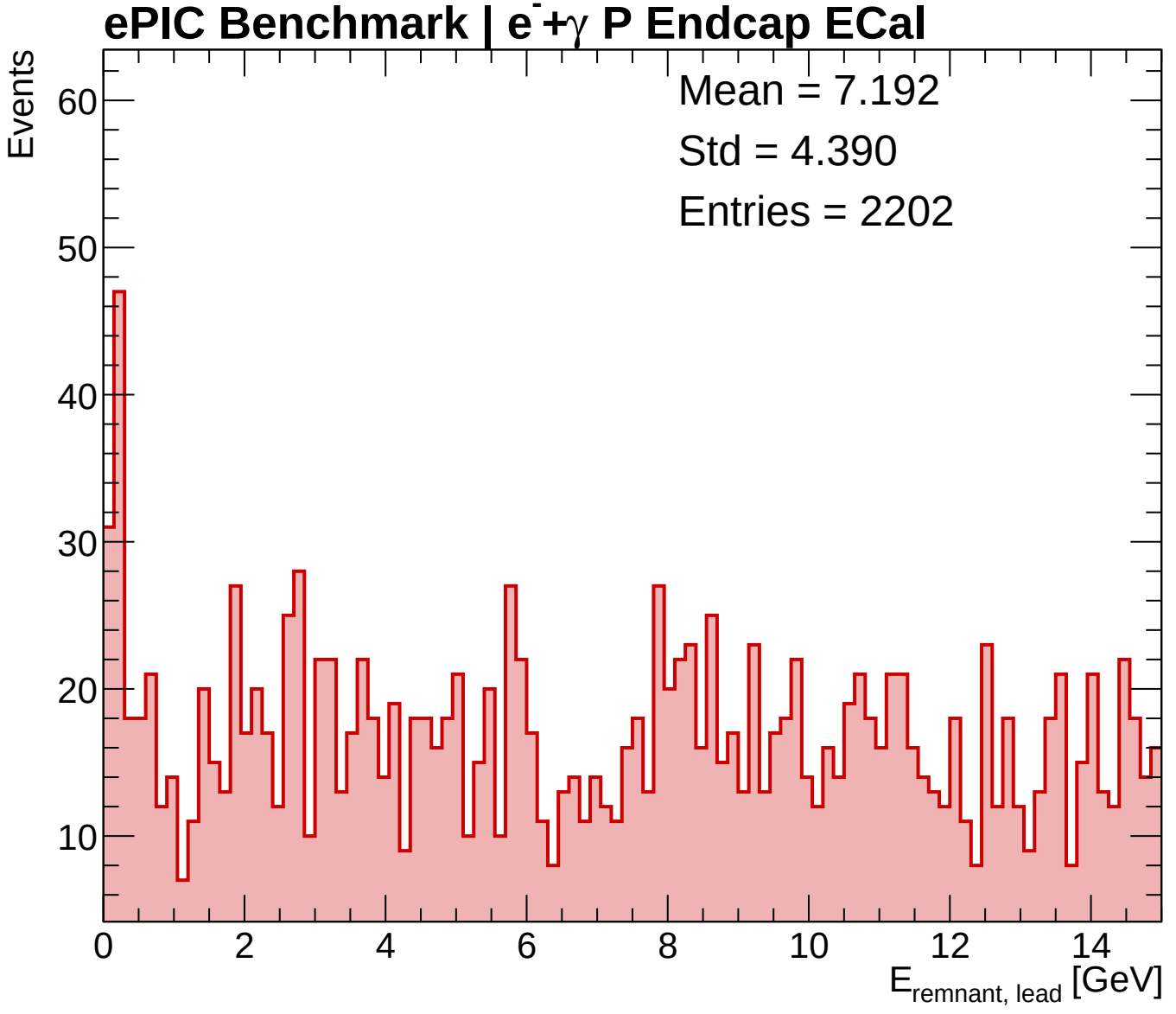

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
0
10
20
30
40
50
60
70
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)

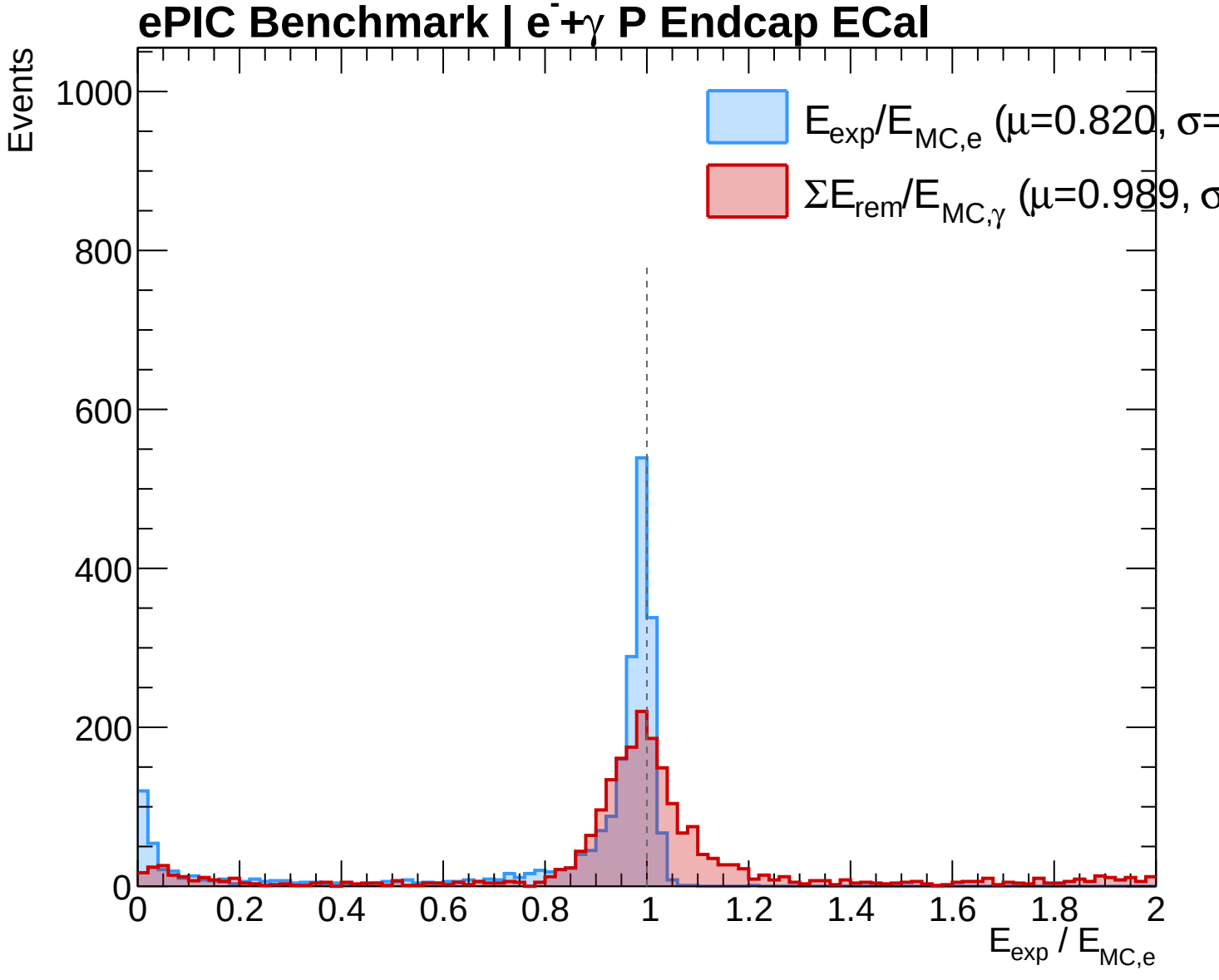

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
10000
8000
6000
4000
2000
0
0
1
2
3
4
5
6
7
8
9
N expected clusters

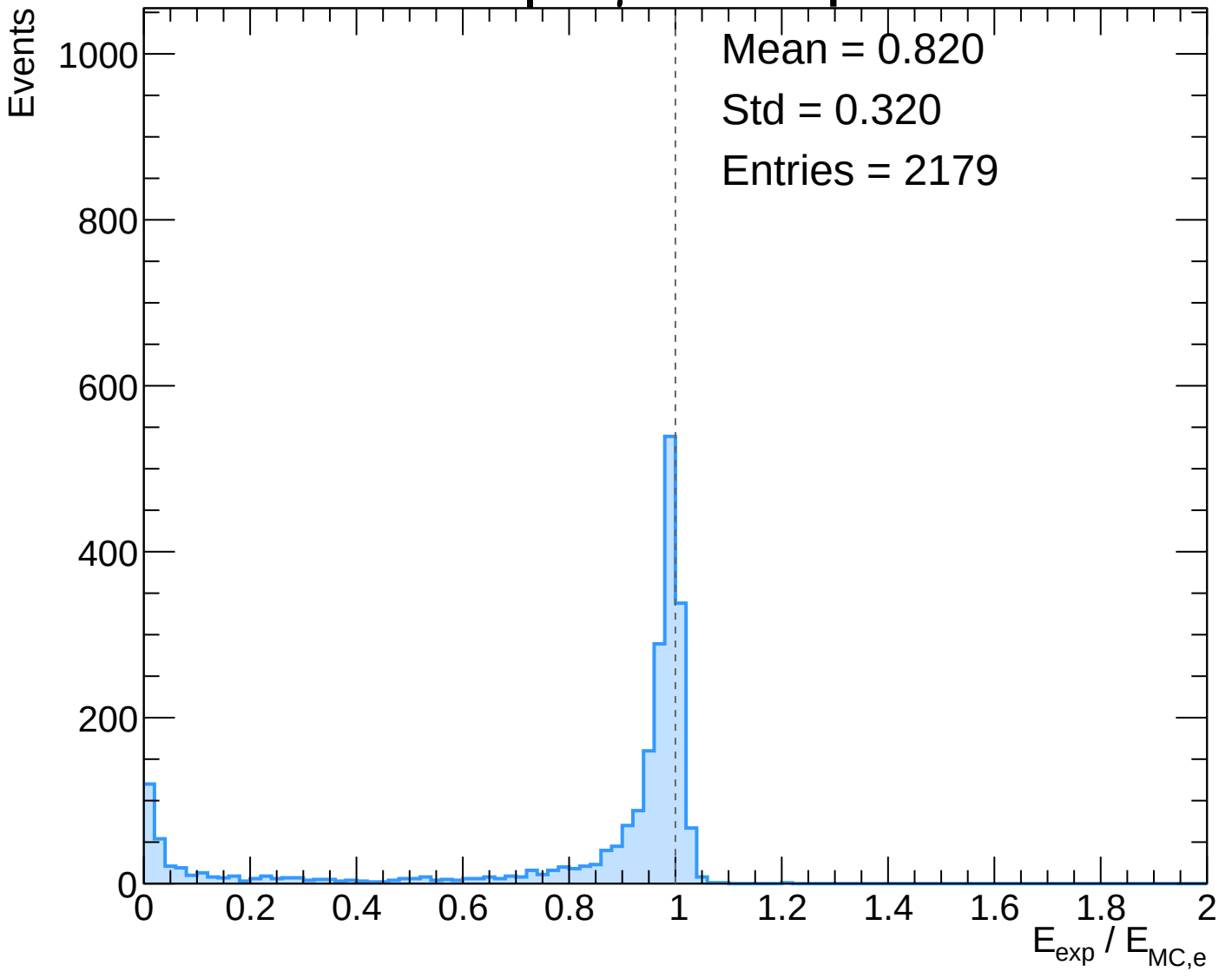

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
10000
8000
6000
4000
2000
0
0
1
2
3
4
5
6
7
8
9
N remnant clusters

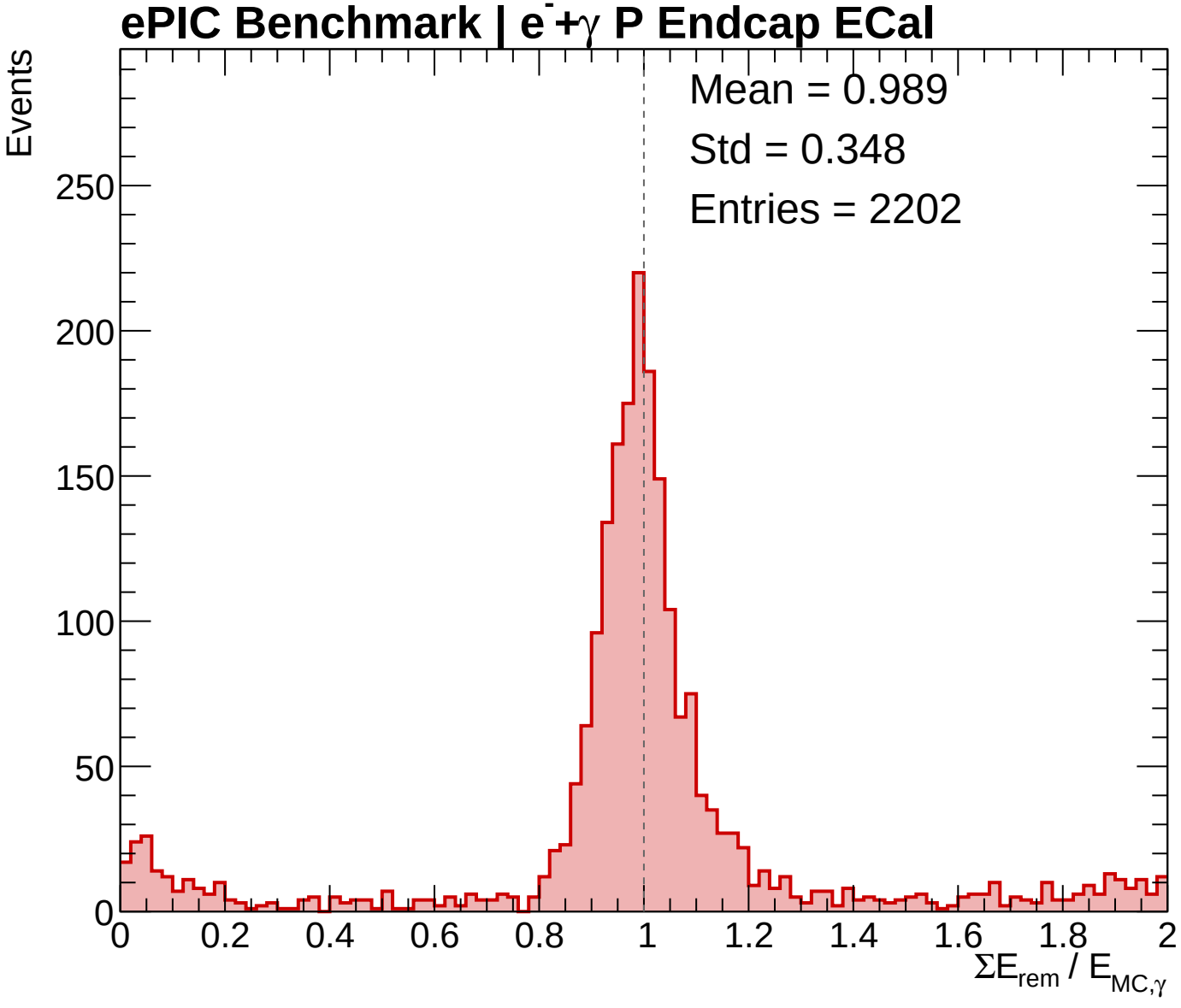

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Mean = 6.583
Std = 4.571
Entries = 2179
120
100
80
60
40
20
0
2
4
6
8
10
12
14
E_expected [GeV]

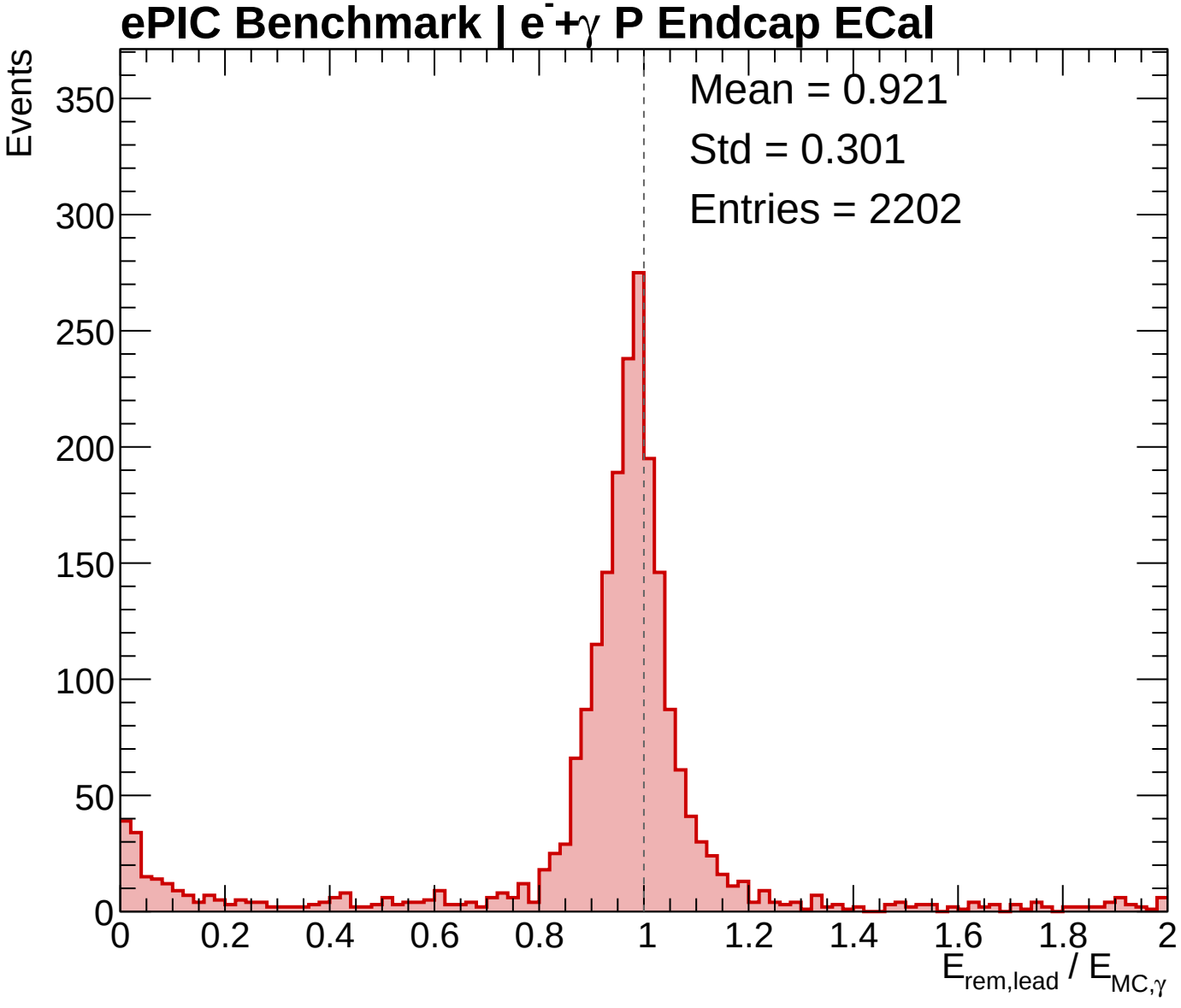

ePIC Benchmark | e⁻+γ P Endcap ECal
Mean = 7.292
Std = 4.297
Entries = 2202
Events
35
30
25
20
15
10
5
0
2
4
6
8
10
12
14
ΣE_remnant [GeV]

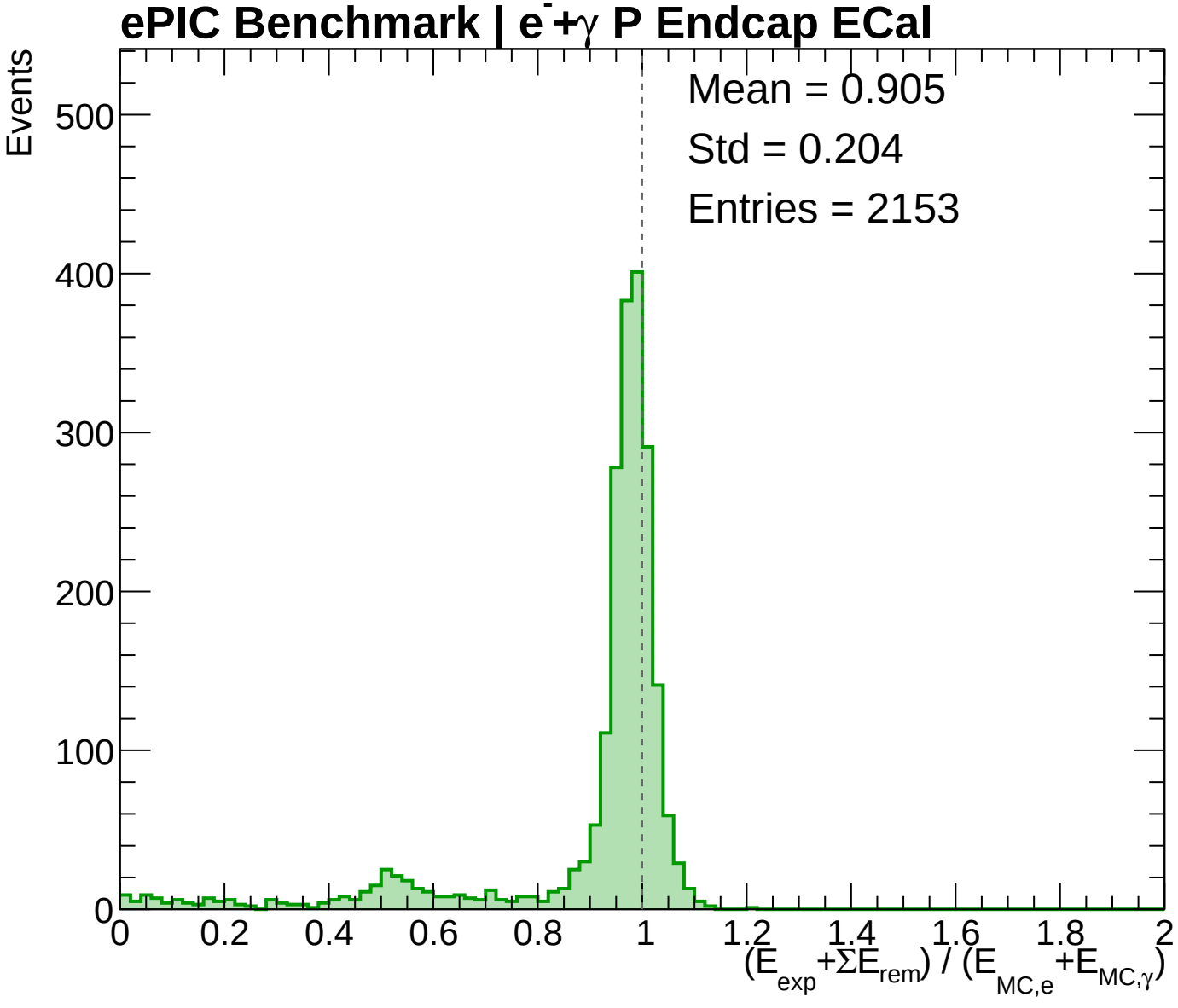

ePIC Benchmark | e⁻+γ P Endcap ECal
Mean = 7.192
Std = 4.390
Entries = 2202
Events
60
50
40
30
20
10
0
2
4
6
8
10
12
14
E remnant, lead [GeV]

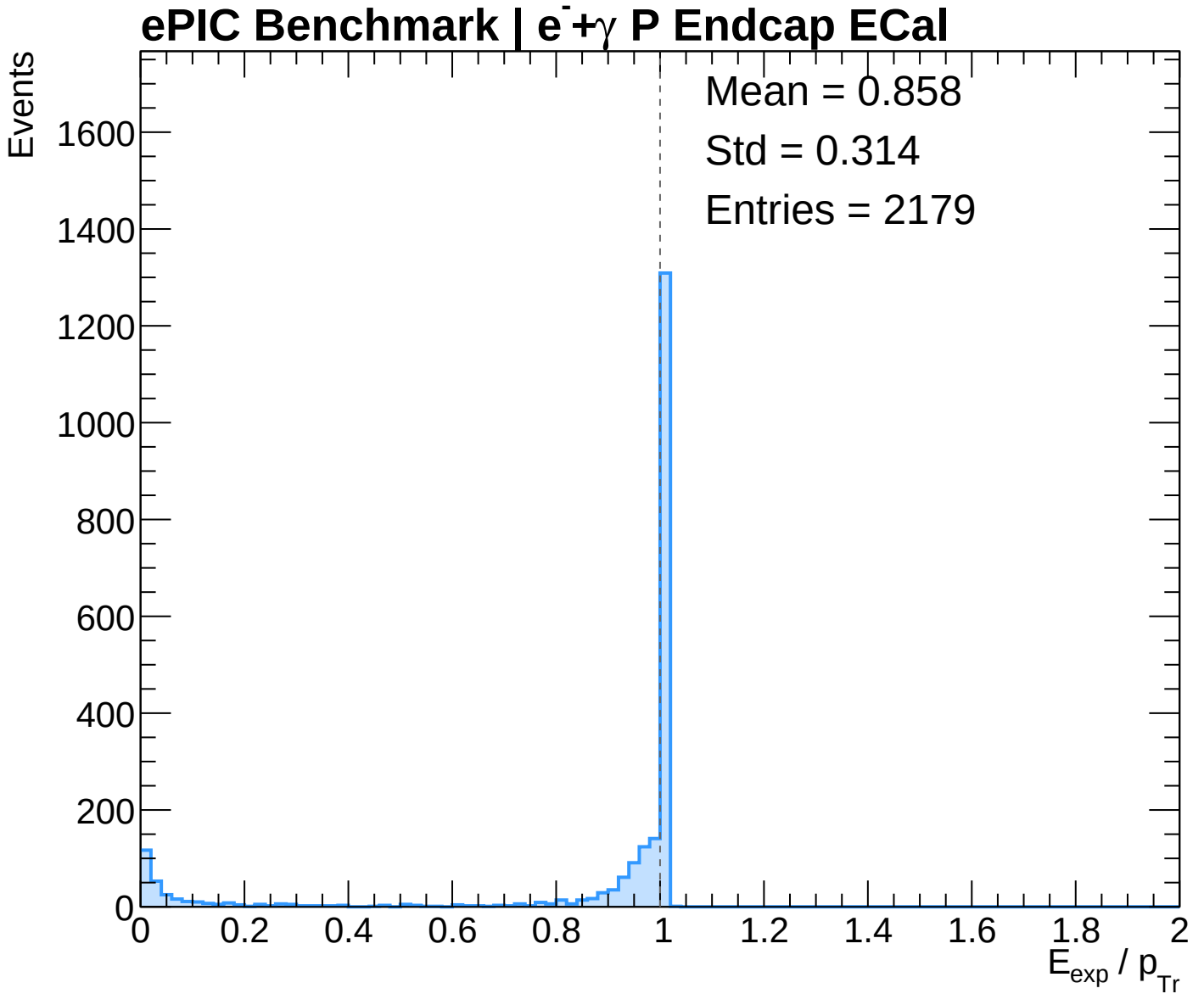

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / E_MC,e
E_exp/E_MC,e (μ=0.820, σ=
ΣE_rem/E_MC,γ (μ=0.989, σ

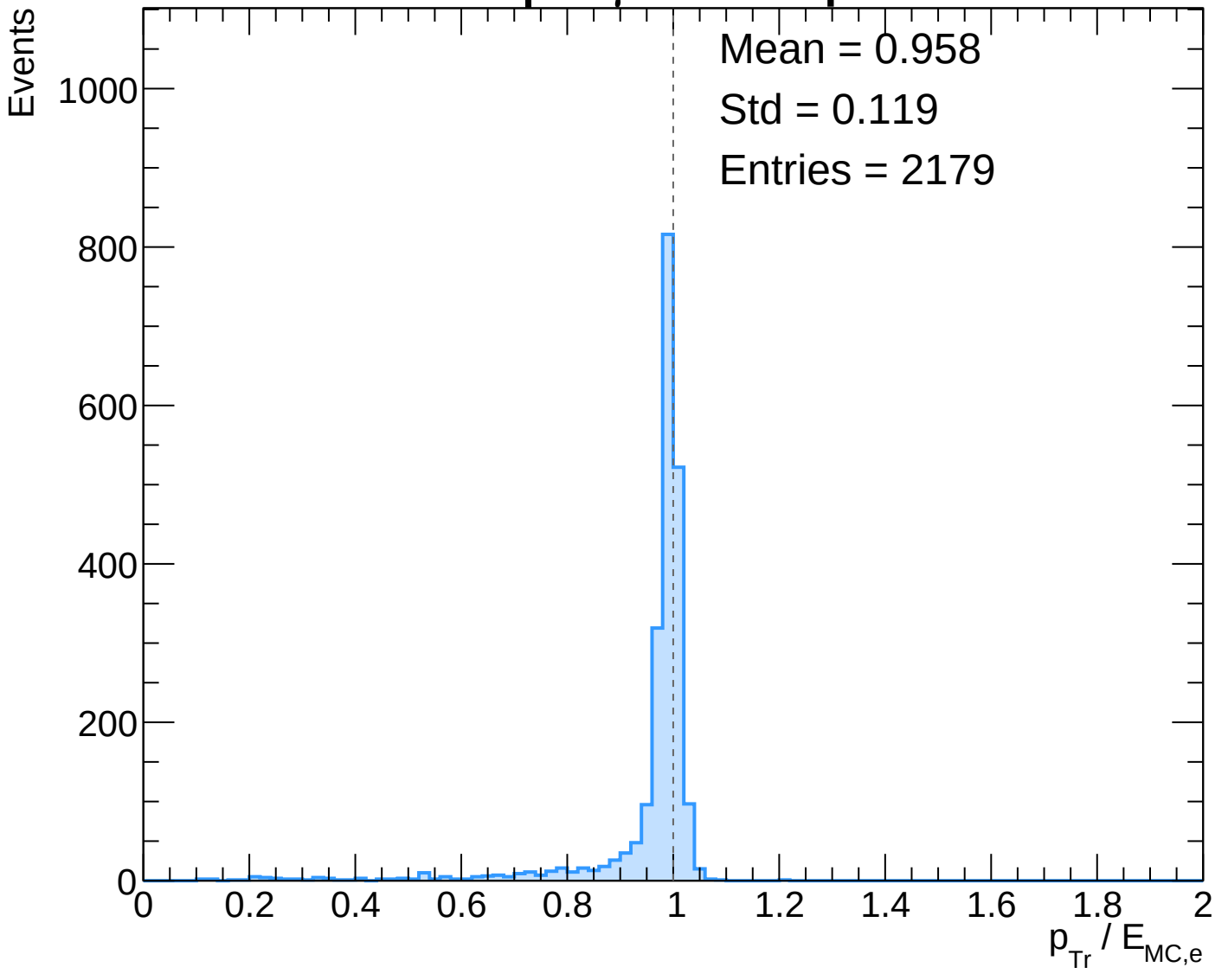

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Mean = 0.820
Std = 0.320
Entries = 2179
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / E_MC,e

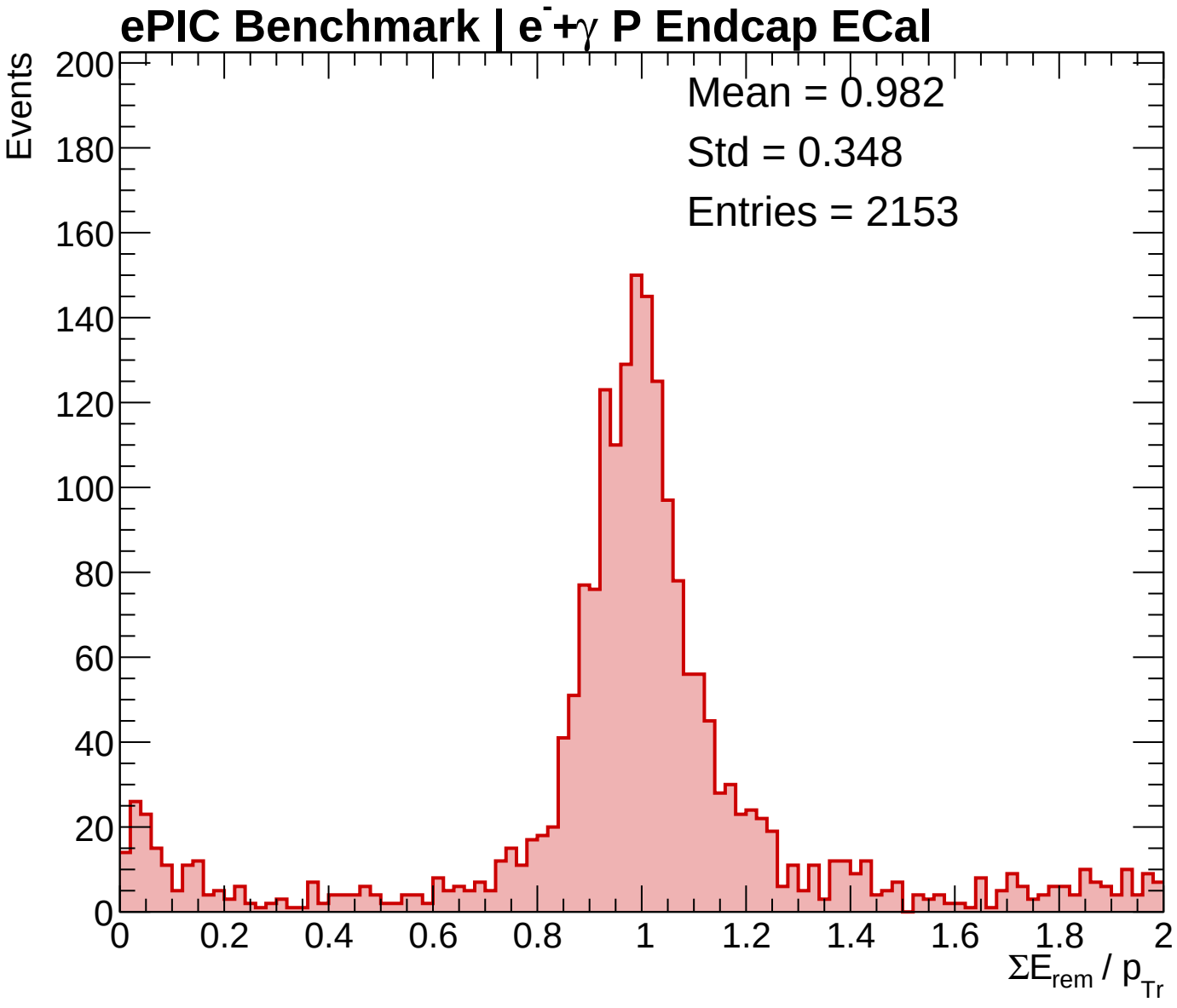

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Mean = 0.989
Std = 0.348
Entries = 2202
0
50
100
150
200
250
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΣE_rem / E_MC,γ

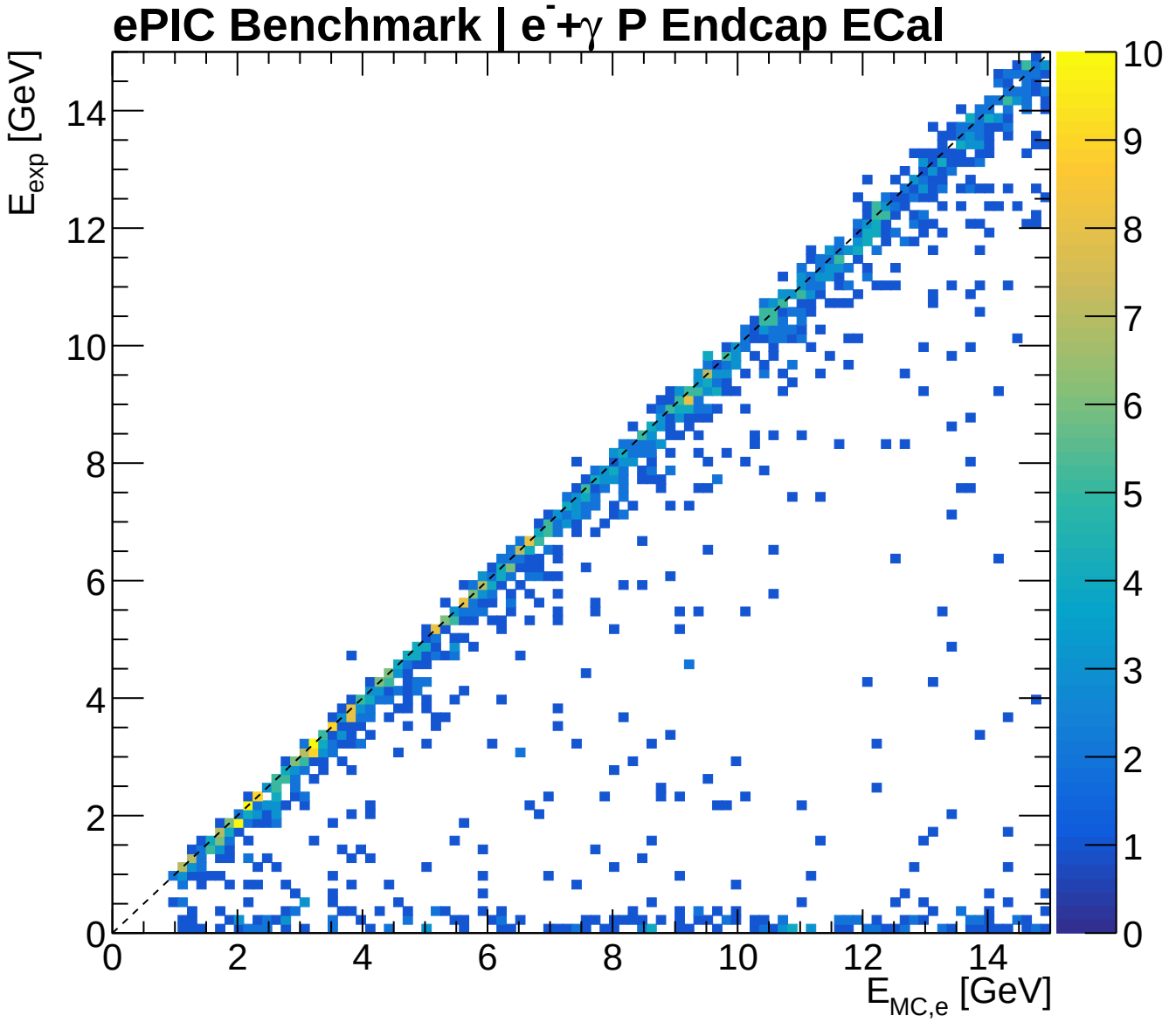

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Mean = 0.921
Std = 0.301
Entries = 2202
0
50
100
150
200
250
300
350
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_rem,lead / E_MC,γ

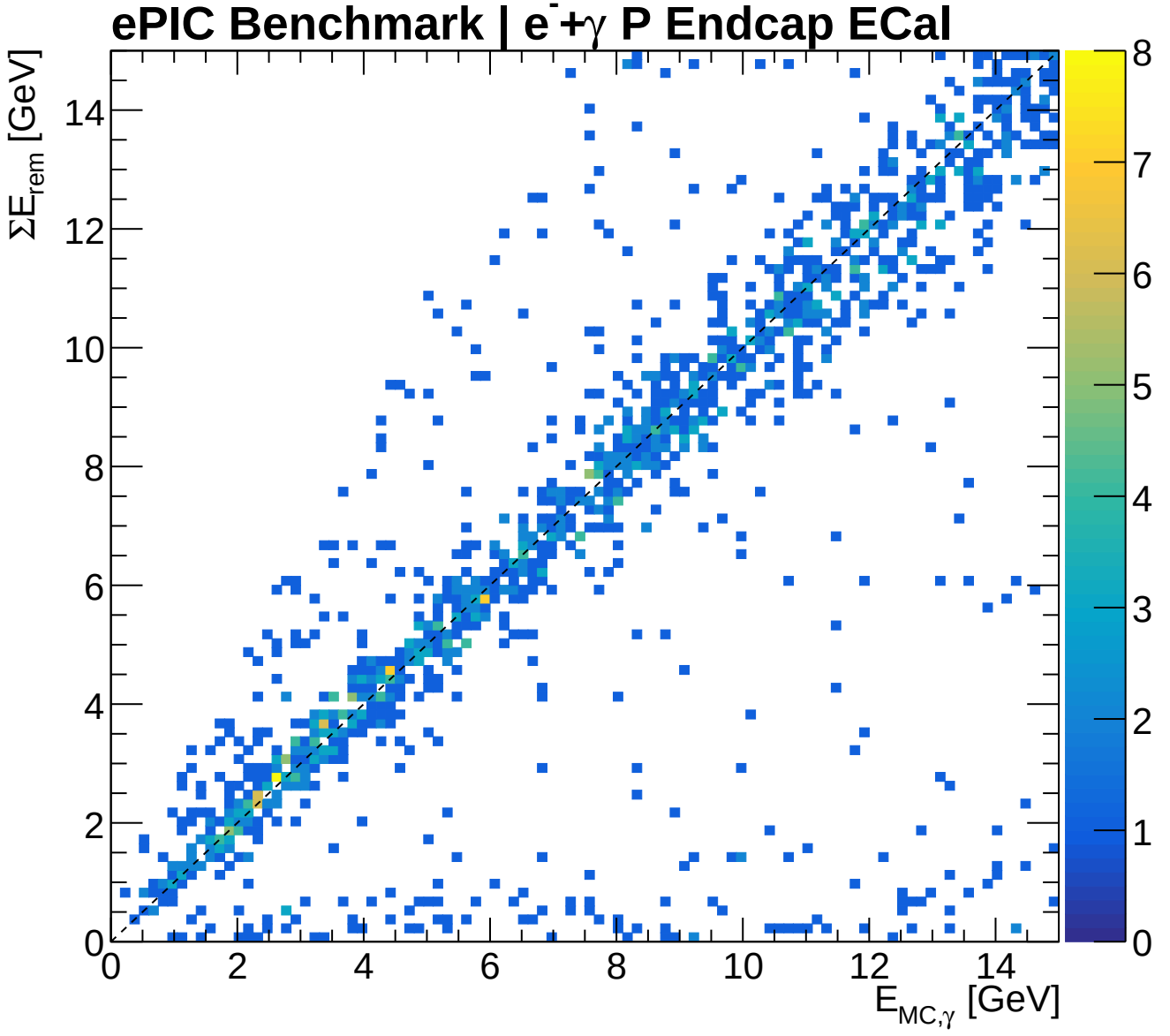

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Mean = 0.905
Std = 0.204
Entries = 2153
0
100
200
300
400
500
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_exp+ΣE_rem) / (E_MC,e+E_MC,γ)

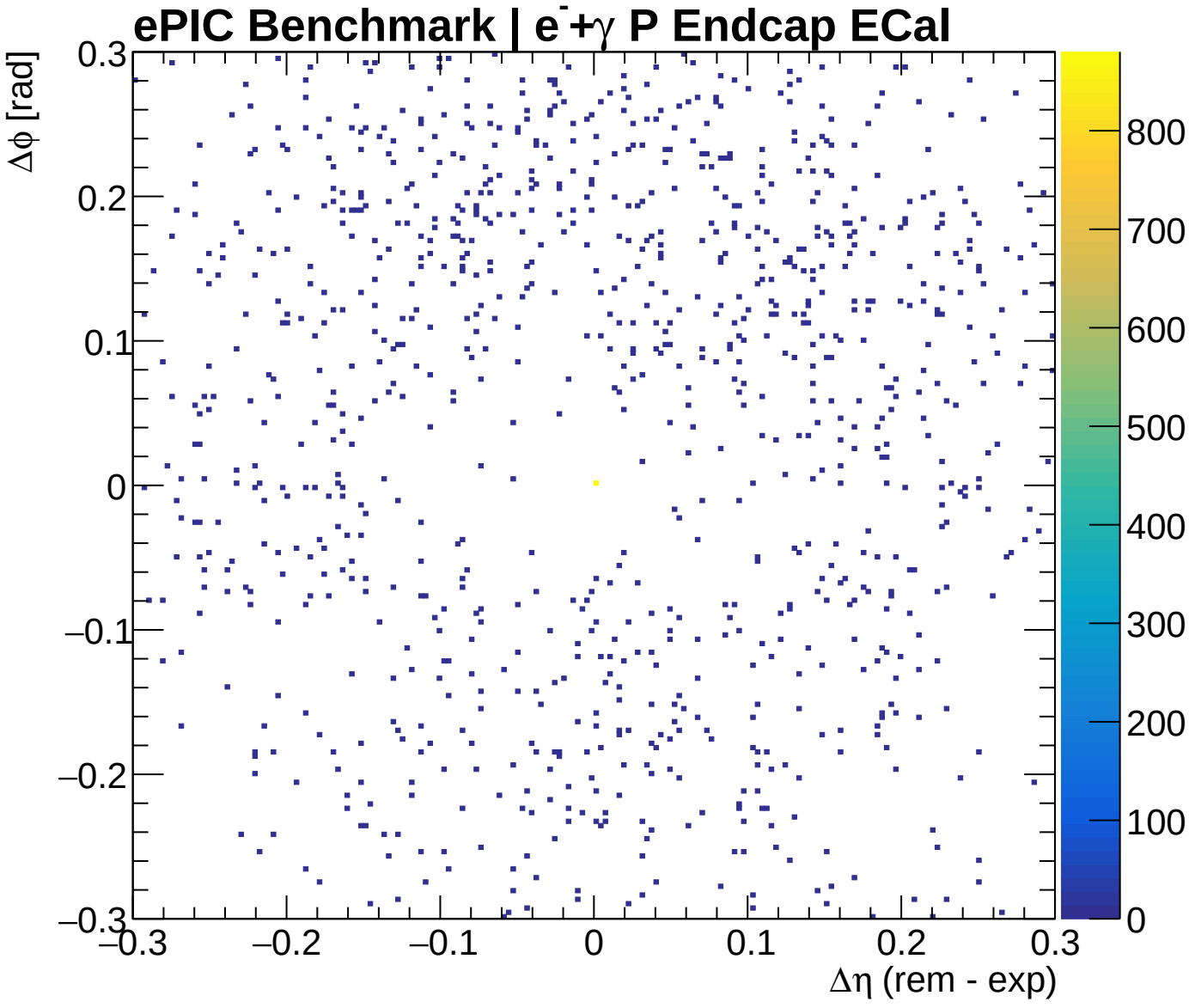

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Mean = 0.858
Std = 0.314
Entries = 2179
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / p_Tr

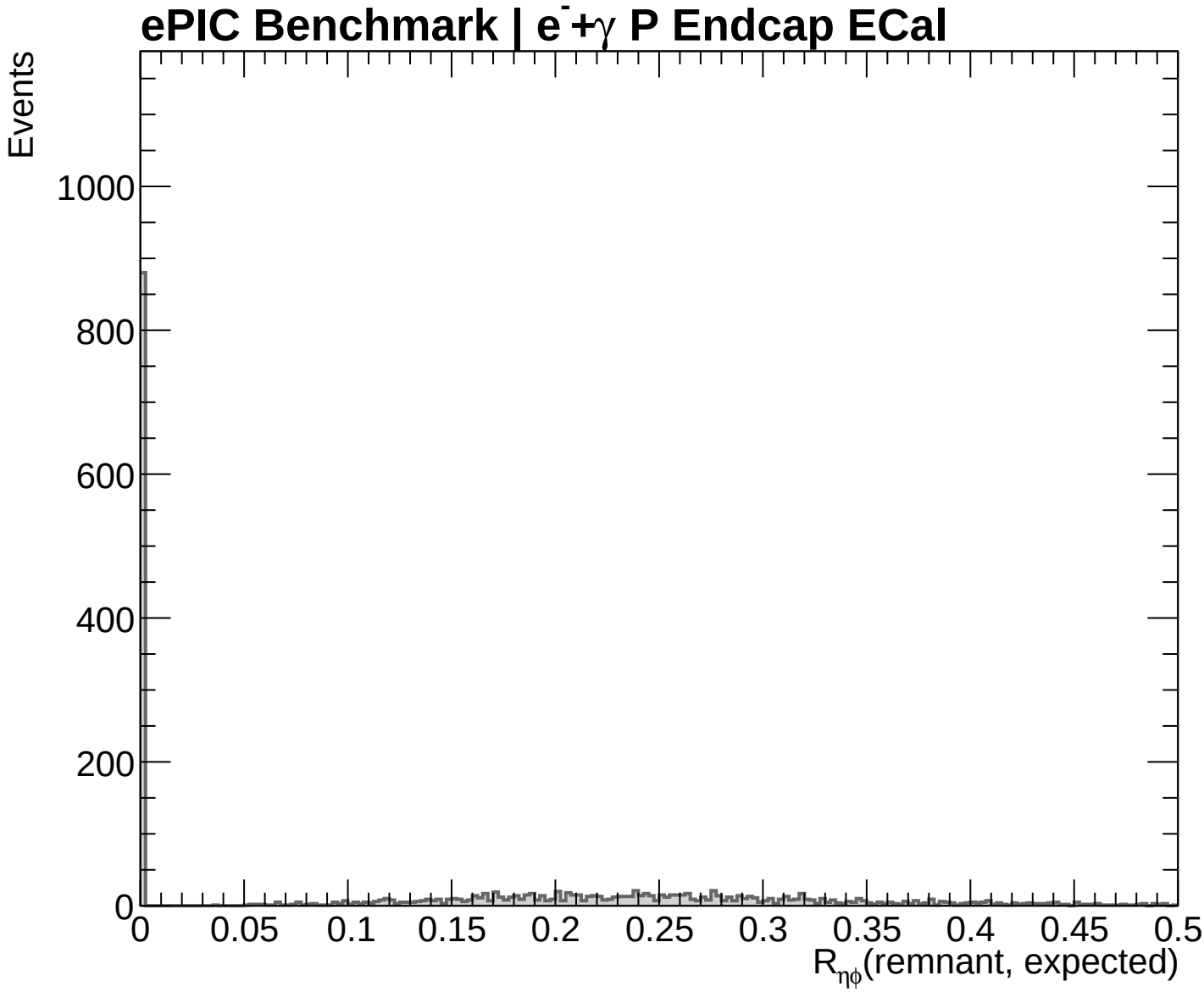

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Mean = 0.958
Std = 0.119
Entries = 2179
0
200
400
600
800
1000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
p_Tr / E_MC,e

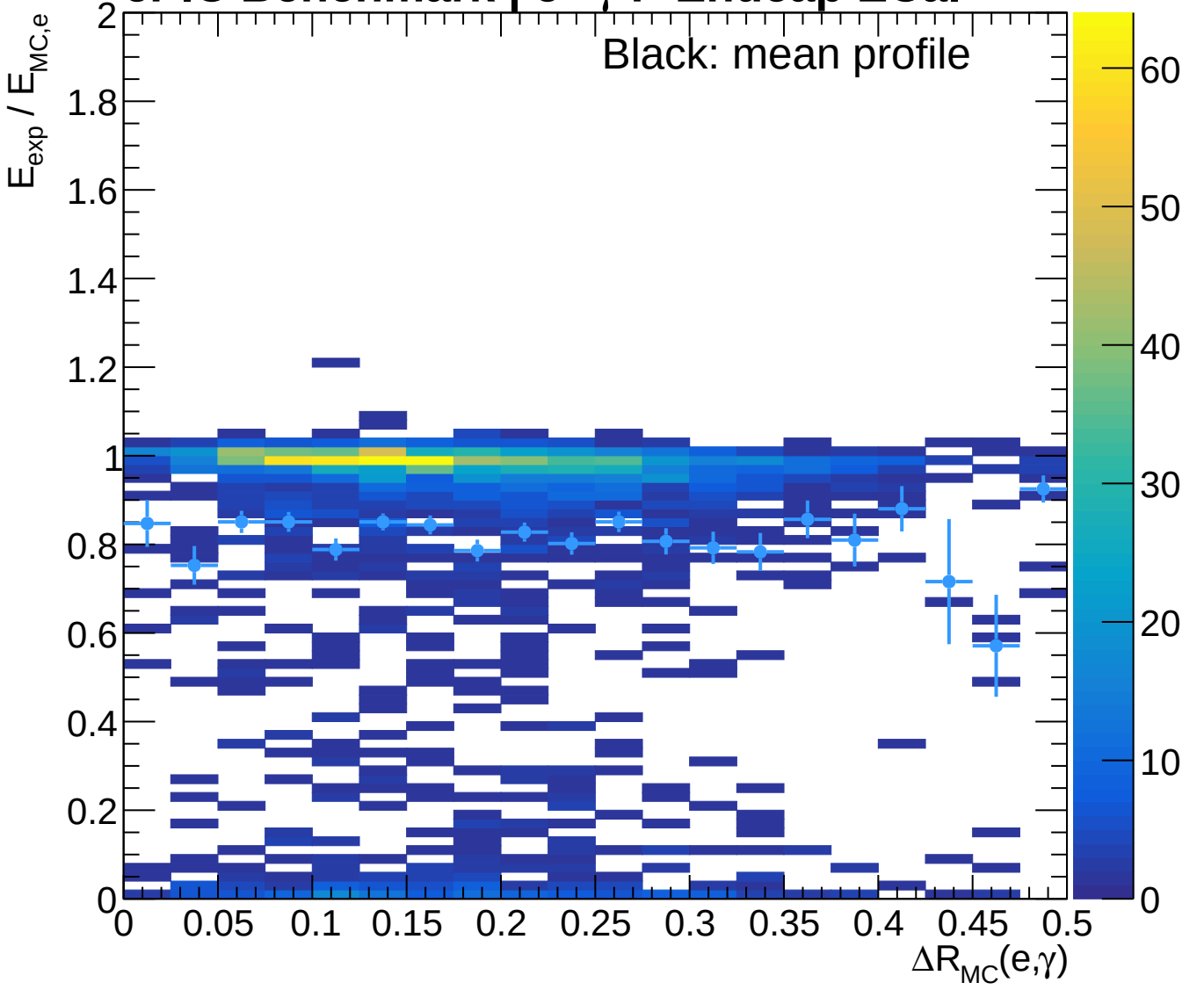

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Mean = 0.982
Std = 0.348
Entries = 2153
0
20
40
60
80
100
120
140
160
180
200
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΣE_rem / p_Tr

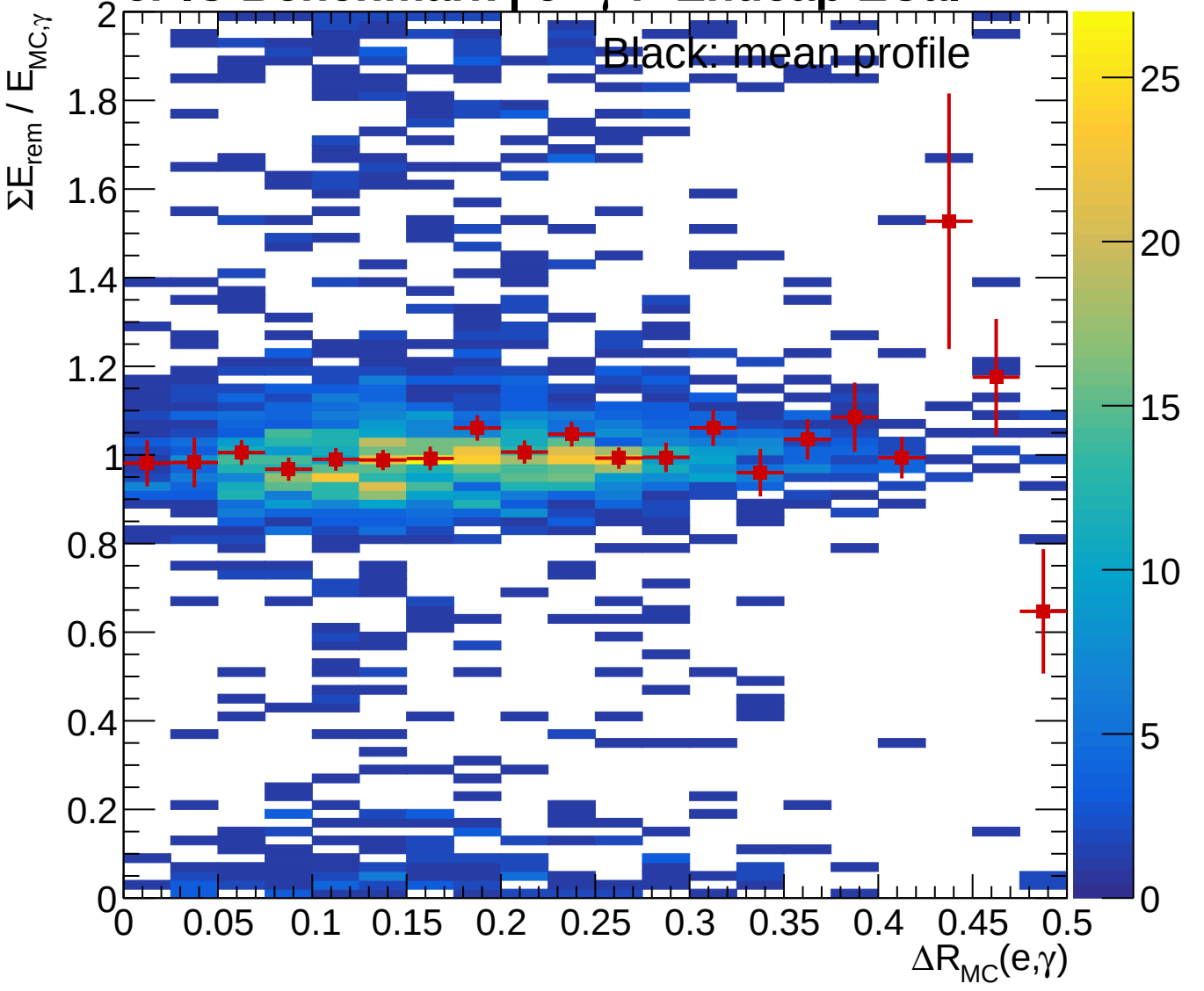

ePIC Benchmark | e⁻+γ P Endcap ECal
E_exp [GeV]
E_MC,e [GeV]
0
2
4
6
8
10
12
14

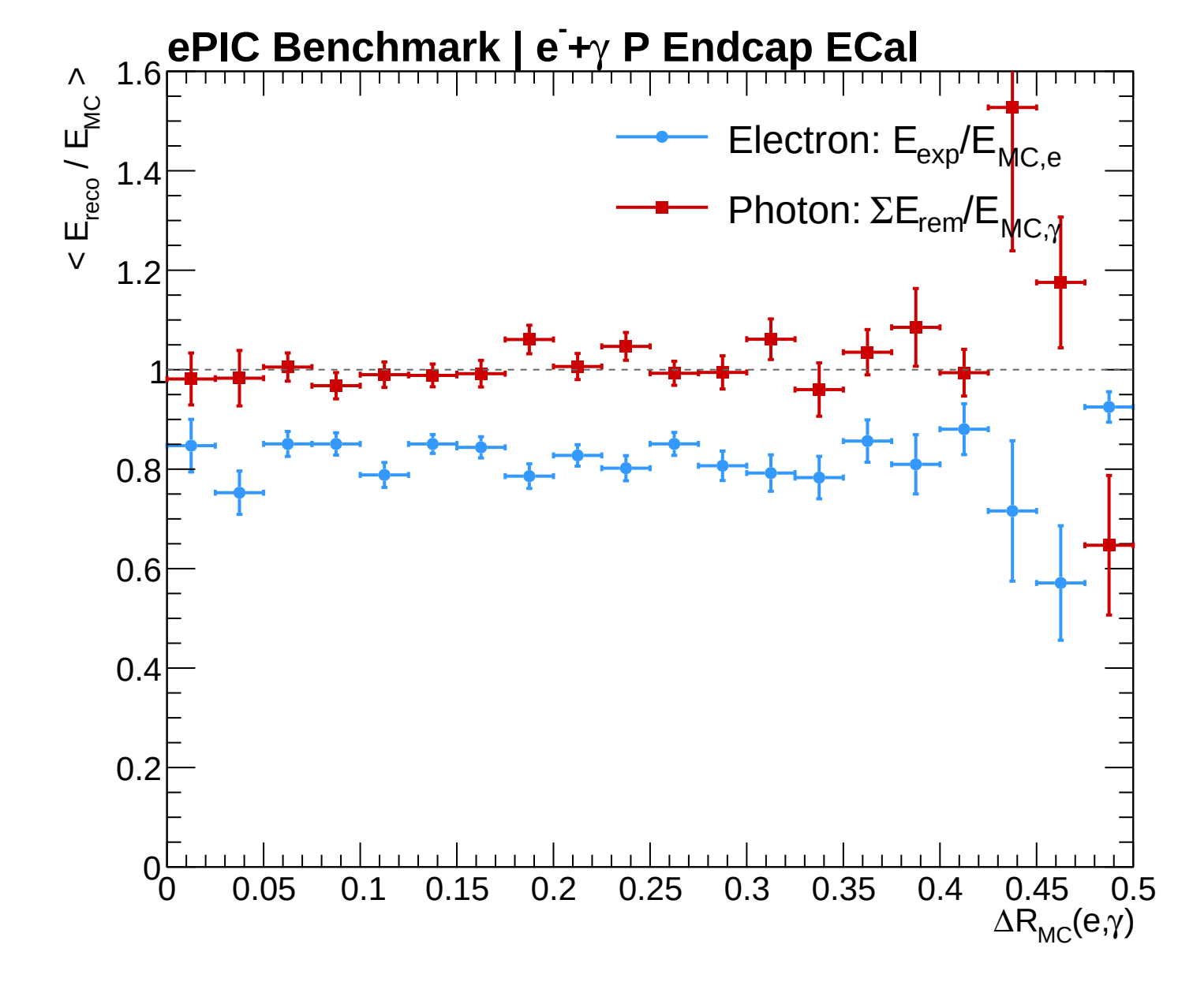

ePIC Benchmark | e⁻+γ P Endcap ECal
ΣE_rem [GeV]
E_MC,γ [GeV]
0
2
4
6
8
10
12
14

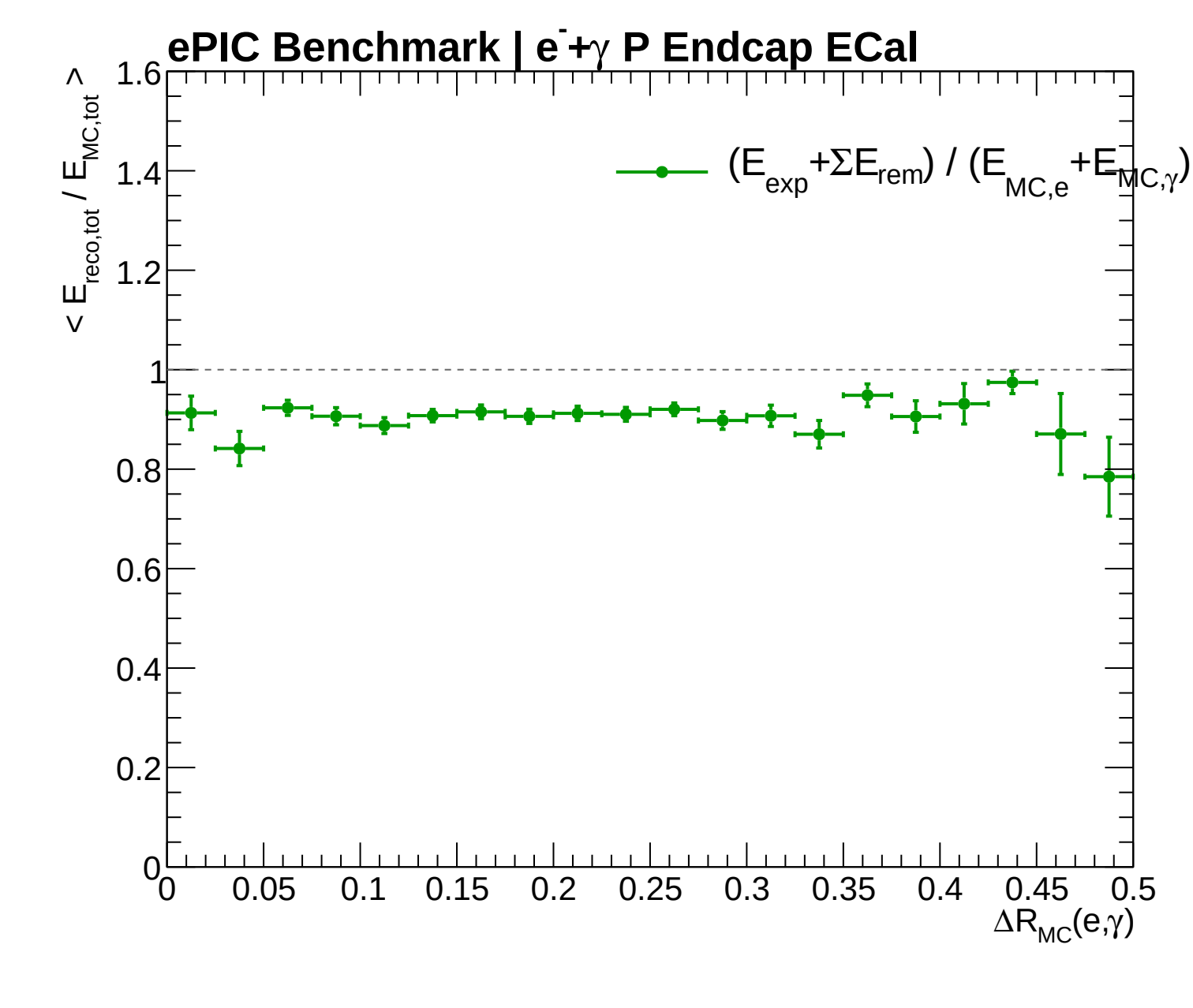

ePIC Benchmark | e⁻+γ P Endcap ECal
Δϕ [rad]
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.3
−0.2
−0.1
0
0.1
0.2
0.3
Δη (rem - exp)
800
700
600
500
400
300
200
100
0

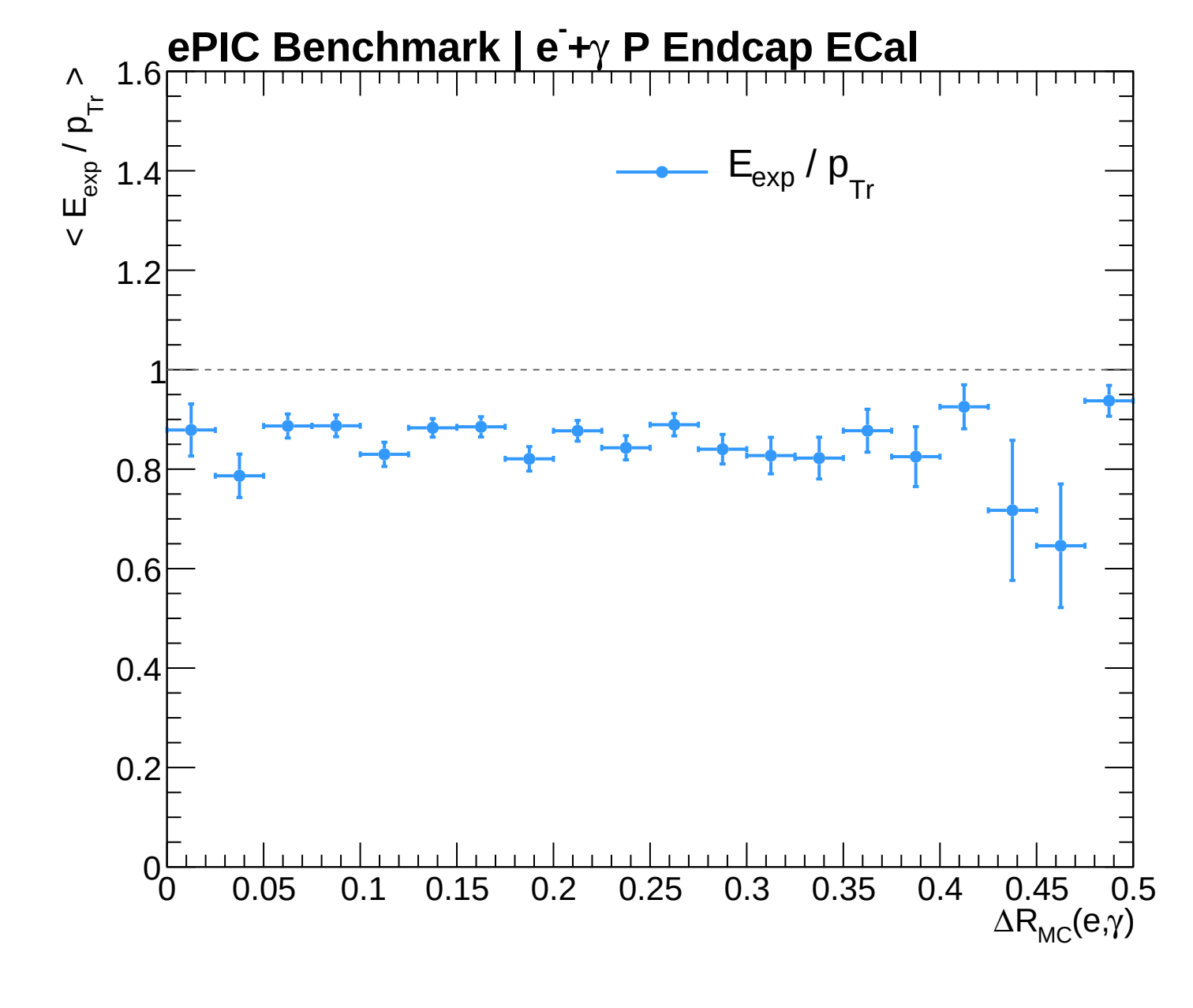

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
1000
800
600
400
200
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
R_ηφ(remnant, expected)

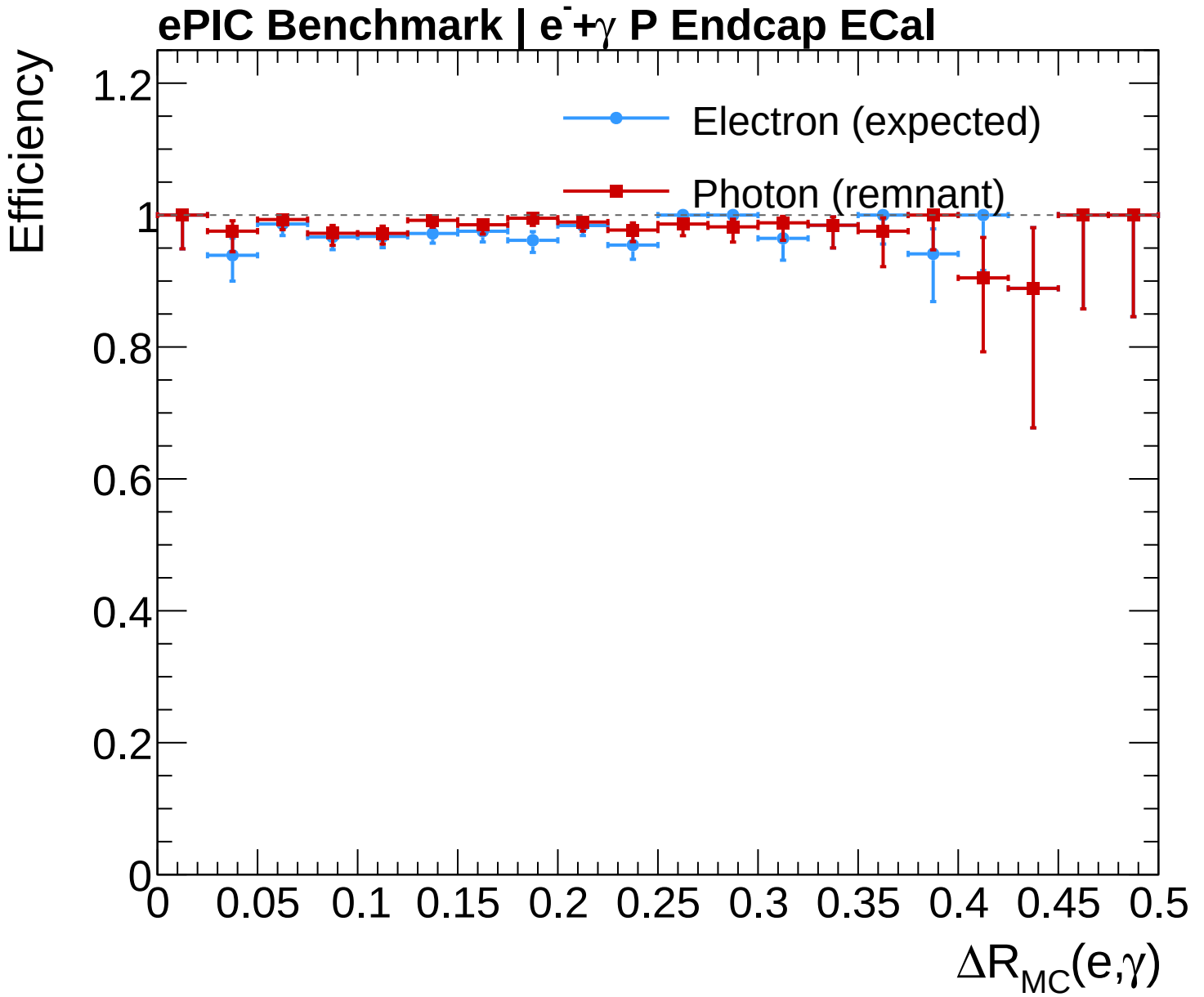

ePIC Benchmark | e⁻+γ P Endcap ECal
Black: mean profile
E_exp / E_MC,e
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
10
20
30
40
50
60

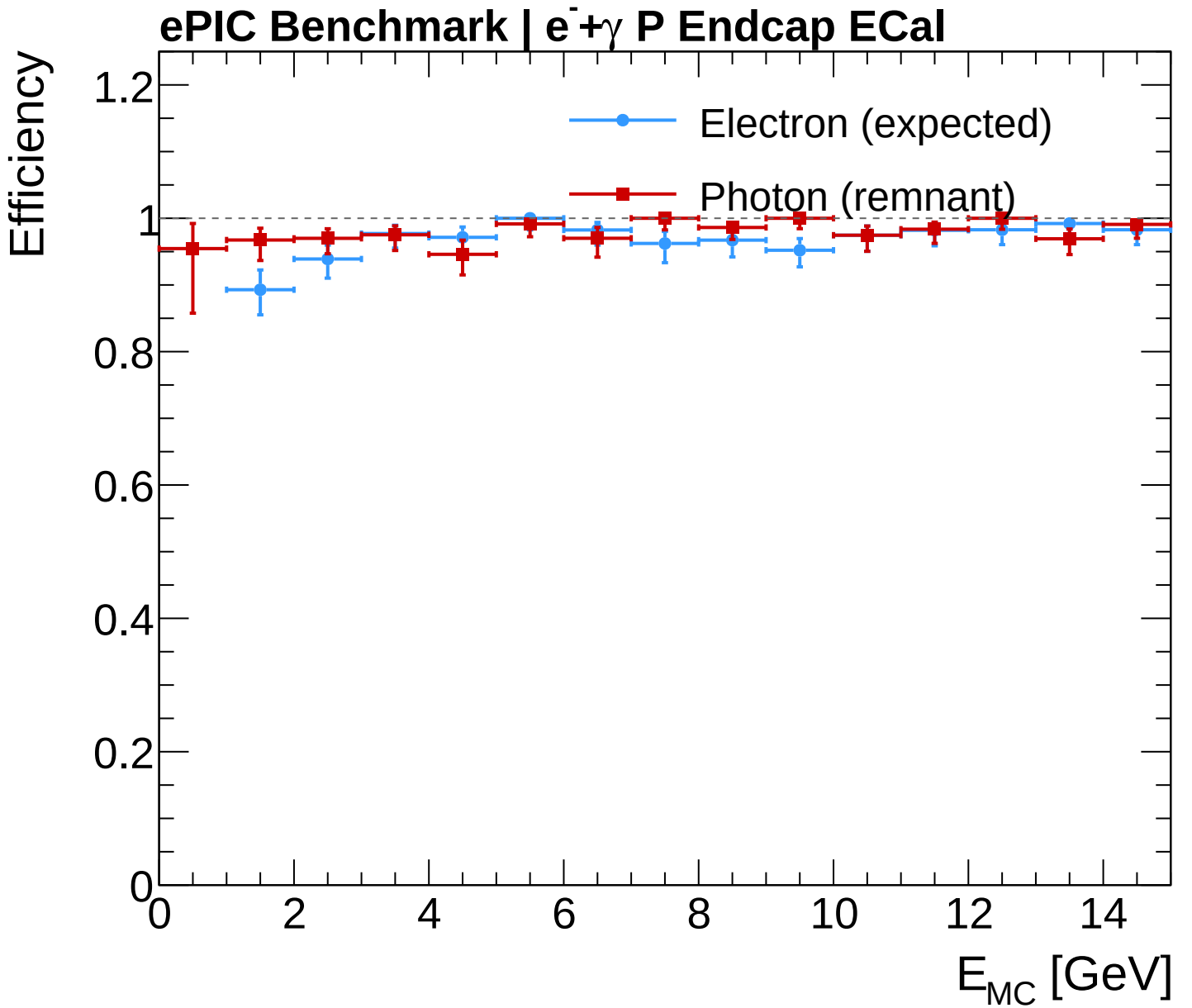

ePIC Benchmark | e⁻+γ P Endcap ECal
Black: mean profile
ΣE_rem / E_MC,γ
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
5
10
15
20
25

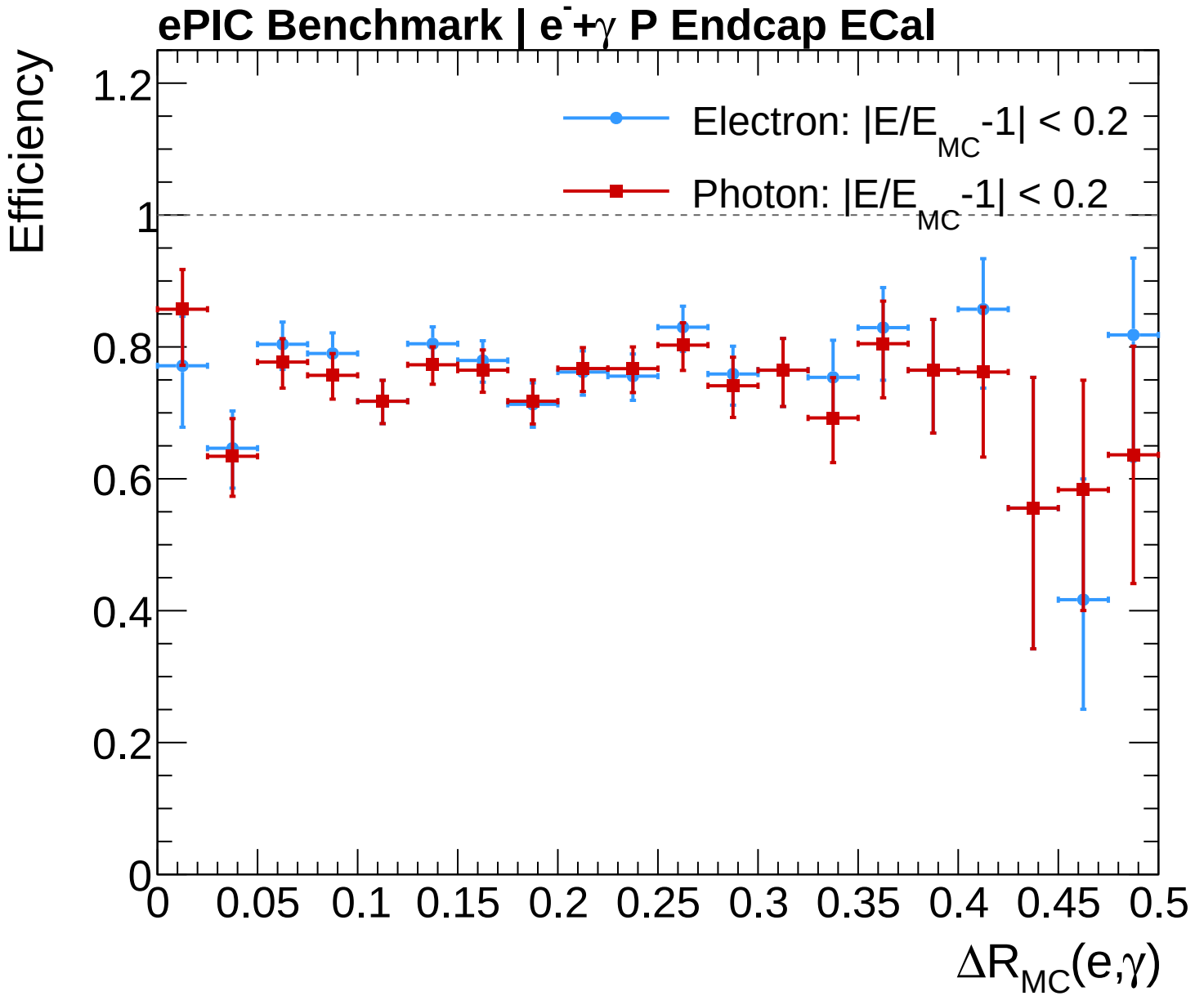

ePIC Benchmark | e⁻+γ P Endcap ECal
< E_reco / E_MC >
Electron: E_exp/E_MC,e
Photon: ΣE_rem/E_MC,γ
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5


ePIC Benchmark | e⁻+γ P Endcap ECal
< E_reco,tot / E_MC,tot >
(E_exp+ΣE_rem) / (E_MC,e+E_MC,γ)
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5


ePIC Benchmark | e⁻+γ P Endcap ECal
< E_exp / p_Tr >
E_exp / p_Tr
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5


ePIC Benchmark | e⁻+γ P Endcap ECal
Efficiency
Electron (expected)
Photon (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ P Endcap ECal
Efficiency
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
E_MC [GeV]
Electron (expected)
Photon (remnant)


ePIC Benchmark | e⁻+γ P Endcap ECal
Efficiency
Electron: |E/E_MC -1| < 0.2
Photon: |E/E_MC -1| < 0.2
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)